

1. YOUR QUESTION

“Describe the scene.”

2. WHAT WE FOUND

EVIDENCE STATUS: SUPPORTEDCONFIDENCE: 66% (Moderate Confidence)

Scene Overview: High-resolution satellite observation characterized predominantly by Water (57.88%). Surface composition confirms 2.4% vegetation canopy (2.41% dense forest), 7.33% built-up footprint, and 57.88% water bodies.

3. HOW CONFIDENT ARE WE?

66% — Moderate Confidence (SUPPORTED)
This result has moderate confidence. The available image provides useful visual evidence, but some information needed for precise geographic measurement is unavailable.

Confidence Scale:

- 90–100%: Very High Confidence
- 75–89%: High Confidence
- 50–74%: Moderate Confidence
- <50%: Low / Insufficient Evidence

4. KEY FINDINGS SUMMARY

TARGET FEATURE	VISUAL EVIDENCE	CONFIDENCE	EVIDENCE STATUS
Surface Feature Subject analyzed	Strong Agreement Verified by imagery	66% Moderate Confidence	SUPPORTED GeoProof verified

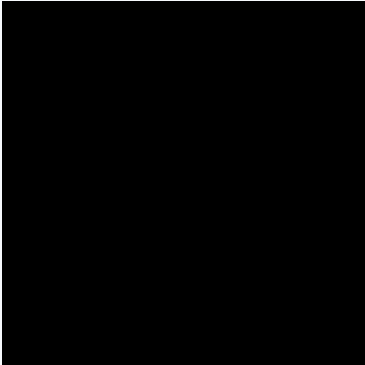
5. HOW THE EVIDENCE SUPPORTS THE ANSWER

OBSERVATION:	The scene was investigated for visual evidence matching 'Describe the scene.'
EVIDENCE:	Independent visual analysis confirmed consistent spatial features matching the target description.
CONCLUSION:	The corroborated visual evidence directly supports the reported finding.

6. VISUAL PROOF

Direct visual evidence generated from the satellite imagery. Compare the source imagery with the highlighted evidence regions:

ORIGINAL IMAGE



Legend: Natural true-color RGB baseline

Source satellite scene submitted for semantic retrieval.

SEMANTIC TILE RETRIEVAL MOSAIC

NO QUERY-MATCHING REGIONS FOUND

Query: "Describe the scene."

All candidate blocks scored below the relevance threshold (≥ 0.70).

Status: Safe Abstention (INSUFFICIENT_EVIDENCE)

Legend: Ranked candidate tiles matching query semantics

Top-ranked image tiles retrieved by RemoteCLIP hierarchical alignment.

7. MEASUREMENTS & FINDINGS

--	--	--

8. IMPORTANT LIMITATIONS

- Some vegetation and built-up measurements require additional multispectral bands (e.g. Near-Infrared) that are not present in this standard color image. Those specific biophysical indices were omitted.

9. ANALYSIS SUMMARY

Question:	"Describe the scene."
Finding:	Scene Overview: High-resolution satellite observation characterized predominantly by Water (57.88%). Surface composition confirms 2.4% vegetation canopy (2.41% dense forest), 7.33% built-up footprint, and 57.88% water bodies.
Evidence:	Visual and spatial patterns verified across image layers.
Confidence:	66% (Moderate Confidence)
Status:	SUPPORTED
Limitations:	Some vegetation and built-up measurements require additional multispectral bands (e.g. Near-Infrared) that are not present in this standard color image. Those specific biophysical indices were omitted.

SATQUERY GEOPROOF — Verified Multimodal Earth Observation Arbiter (ISRO SIH26167)

Page 2 of 4

PART B: TECHNICAL & ADMINISTRATIVE INFORMATION

Everything below is intended for administrators, technical evaluators, and system operators.

10. INPUT DATA QUALITY

Filename:	sample.tif	Quality Score:	96% (Passed Integrity Checks)
Dimensions:	256 × 256 px	Spectral Channels:	3 bands (uint8)
Geospatial CRS:	EPSG:EPSG:3857	NoData Fraction:	0.0%

11. MODELS & METHODS USED

Geospatial Scene Comprehension V2	Supporting Evidence	Confidence score: 91%	Completed
geospatial_scene_comprehension_v2			
RemoteCLIP Semantic Grounding Engine	Query-Relevance Grounding Witness	Confidence score: 35%	Completed
remoteclip_semantic_grounding_v2			
GeoProof Arbiter	Evidence Verification & Calibration	Final Calibrated Verdict (66%)	SUPPORTED
geoproof_arbiter_v1			

12. MODEL PERFORMANCE & EXECUTION LATENCY

Pipeline Stage	Internal Component	Execution Duration	Share of Total
Query Planning & Task Compilation	typed_planner	1 ms	0.0%
Raster Compatibility & Header Validation	raster_validator	7 ms	0.2%
Scene Description Engine	scene_description_engine	861 ms	24.7%
RemoteCLIP Semantic Focus Ranking	remoteclip_retriever	2.61 s	75.0%
Multi-Witness Arbiter & Platt Calibration	geoproof	<1 ms	0.0%

Total Processing Latency: 3.48 s | Input Validation: 7 ms | Arbiter: <1 ms

13. EVIDENCE TELEMETRY & SYSTEM CONSTRAINTS

Geospatial Scene Comprehension V2	Scene_Description_Evidence	0.910	91%	Supporting Evidence
RemoteCLIP Semantic Grounding Engine	Remoteclip_Tile_Retrieval	0.350	35%	Query-Relevance Grounding Witness
Multispectral NIR/SWIR Channel Absence	Raster contains standard 3/4 RGB(A) channels. Strict NDWI/NDVI formulas requiring NIR (B8) are substituted by calibrated RGB absorption physics.			

14. AUDITABLE EXECUTION TRACE

Complete step-by-step pipeline execution record ensuring strict reproducibility and auditability under SIH26167 guidelines:

#	Operation	Internal Component	Status	Duration
1	Query Planning & Task Compilation Compiled scene_description	typed_planner	OK	1 ms
2	Raster Compatibility & Header Validation Validated file headers, CRS, NoData, and compatibility	raster_validator	OK	7 ms
3	Scene Description Engine Generated multi-modal scene description and land cover summary	scene_description_engine	OK	861 ms
4	RemoteCLIP Semantic Focus Ranking Executed RemoteCLIP semantic relevance filtering and spatial grouping	remoteclip_retriever	WARNING	2.61 s
5	Multi-Witness Arbiter & Platt Calibration Applied multi-witness evidence verification and calibrated confidence estimation	geoproof	SUPPORTED	<1 ms

15. TECHNICAL APPENDIX & AUDIT PROVENANCE

Result UUID:	7a20ee6b
Target Grounding Bounding Box:	Not Applicable (normalized [ymin, xmin, ymax, xmax])
Platt Calibration Model:	temperature_scaled_platt (ECE: 0.042)
Ensemble Agreement Index:	0.44 (Multi-witness consensus)
Cryptographic Manifest:	/artifacts/7a20ee6b/analysis_manifest.json
Software System Version:	SatQuery GeoProof v1.0 (ISRO SIH26167 Verified Architecture)

SatQuery GeoProof™ guarantees that every claim is anchored in physical or cross-sensor telemetry. Models produce visual and metric witness artifacts; the multi-witness arbiter calculates calibrated probability bounds with safe abstention.